

in the US military, which is interested in developing unconventional energy sources to power a soldier's equipment such as radios and electronic gun sights. By harvesting power generated by walking, soldiers of the future won't have to carry as many batteries and will have more room for other supplies such as ammunition and food.

Power to the chip

The silicon chip has been the fundamental building block that has powered the computer revolution. Silicon's properties make it possible for information to be moved quickly via miniaturised transistors. But the silicon revolution is not over yet; it may very well be used as a power source too. Scientists have successfully made fuel-based chip processors that channel chemical reagents to produce power. In one experiment, scientists successfully created a tiny steam engine out of a silicon chip. Instead of electrical impulses carrying data, the chip's channels carried steam. In another experiment, researchers at Sandia National Laboratories made a geared engine on a chip.

Such chip-based fuel processors coupled with miniature methanol or hydrogen-based fuel cells could power portable electronic devices in the future. This has many implications, and could trigger a second silicon revolution. A working power plant on a silicon chip could be used to miniaturise electronics. Scientists speculate that one chip alone could not power a laptop, but say that several of these chips working in tandem could easily power a mobile device. One experiment in Germany indicated that chip-based fuel processors could power laptops for up to 10 hours—a lot longer than the current capability of standard batteries.

Less gives more

PicoDyne, a semiconductor start-up, plans to release a range of chips that consume 50 to 100 times less power than current notebooks. Most of today's chips require 3 volts; Picodyne's chips promise to power high performance electronic gadgets with far lesser power. These chips could save power in a range of equipment, from laptops and PDAs to cell phones and hearing aids.

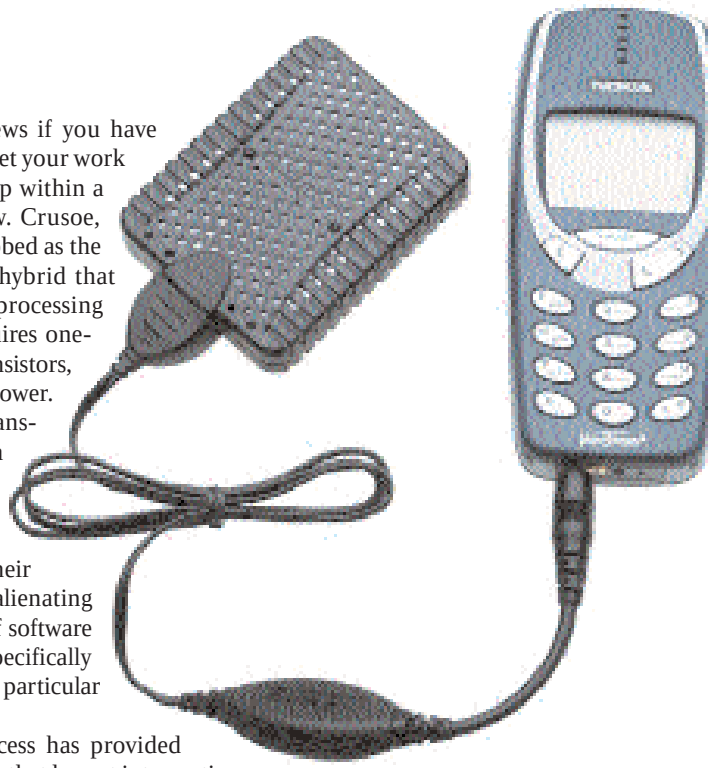
Transmeta was the first to announce Crusoe, a low power chip that is Intel-compatible with one-tenth the power requirements of a Pentium

III. That's good news if you have been struggling to get your work done on your laptop within a three-hour window. Crusoe, which has been dubbed as the software chip, is a hybrid that shifts most of the processing to software. It requires one-fourth as many transistors, and hence less power. Simultaneously, Transmeta has found a way to radically improve the ability of chip designers to make changes in their products without alienating the huge libraries of software that have been specifically written to run on a particular piece of hardware.

Transmeta's success has provided the proof of concept that has set into motion similar projects by IBM, HP, and others. Elbrus, a Moscow-based group has also designed a prototype on similar lines. Its E2K chip is pegged to run faster than Intel's new Itanium chip and still be compatible with all existing Intel-compatible software. A six-year-old IBM project, code-named Daisy, has looked at similar development techniques for the PowerPC chip. Compaq's Wiggins/Redstone project uses a sampling technique, which determines where most of the CPU cycle's time in a program are used, which it then dynamically optimises.

In fact, Intel itself has employed dynamic translation techniques to mimic the Itanium instruction set. The goal of the Intel project is to create a virtual Itanium-compatible chip in the software layer, which can run on Intel's current generation of processors. Software developers will thus be able to write applications for Itanium even before the chip is available. A project, code-named Dynamo, from the HP Labs is being tested at four universities. While a 700 MHz Crusoe chip would work only as fast as a 500 MHz Pentium, HP claims that applications will run faster with Dynamo than without due to code optimisation.

With battery power and power consumption entering new vistas, the days of racing to get the work done before your notebook battery runs out of juice may soon be over. 



Zinc-air batteries can outrun alkaline ones by three times

<http://www>

<http://www.mbi.panasonic.co.jp>

For news and updates on Panasonic's research on Lithium-ion batteries

www.homepower.com

For technical know-how on building a self-powered home

www.realgoods.com

A site for eco-friendly products

www.research.ibm.com/daisy

Information on IBM's Daisy project